

18.600: Lecture 4

Axioms of probability and inclusion-exclusion

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Outline

Axioms of probability

Consequences of axioms

Inclusion exclusion

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Inclusion exclusion

Axioms of probability

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- ▶ Finite additivity: $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$.
- ▶ Countable additivity: $P(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$ if $E_i \cap E_j = \emptyset$ for each pair i and j .

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- ▶ Friend: "Wow... you've beat by suggested price by 50 cents on each deal. Yes, sure! You're a great friend!"
- ▶ Axioms breakdowns are money-making opportunities.

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Intersection notation

- ▶ We will sometimes write AB to denote the event $A \cap B$.

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- ▶ Can we show from the axioms that if S contains finitely many elements $x_1, \dots, x_k$, then the values $(P(\{x_1\}), P(\{x_2\}), \dots, P(\{x_k\}))$ determine the value of $P(A)$ for any $A \subset S$?

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- ▶ What k -tuples of values are consistent with the axioms?

Famous 1982 Tversky-Kahneman study (see wikipedia)

- ▶ People are told “Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations.”

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- ▶ 85 percent chose the second option.
- ▶ Could be correct using neurological/emotional definition. Or a “which story would you believe” interpretation (if witnesses offering more details are considered more credible).
- ▶ But axioms of probability imply that second option cannot be more likely than first.

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Inclusion-exclusion identity

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- ▶ There are some situations in which computing $P(E_1 \cup E_2 \cup \dots \cup E_n)$ is a priori difficult, but it is relatively easy to compute probabilities of *intersections* of any collection of E_i . That is, we can easily compute quantities like $P(E_1 E_3 E_7)$ or $P(E_2 E_3 E_6 E_7 E_8)$.

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- ▶ In these situations, the inclusion-exclusion rule helps us compute unions. It gives us a way to express $P(E_1 \cup E_2 \cup \dots \cup E_n)$ in terms of these intersection probabilities.

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- ▶ More generally,

$$\begin{aligned} P(\cup_{i=1}^n E_i) &= \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} E_{i_2}) + \dots \\ &\quad + (-1)^{(r+1)} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \\ &\quad + \dots + (-1)^{n+1} P(E_1 E_2 \dots E_n). \end{aligned}$$

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- ▶ The notation $\sum_{i_1 < i_2 < \dots < i_r}$ means a sum over all of the $\binom{n}{r}$ subsets of size r of the set $\{1, 2, \dots, n\}$.

Inclusion-exclusion proof idea

- ▶ Consider a region of the Venn diagram contained in exactly $m > 0$ subsets. For example, if $m = 3$ and $n = 8$ we could consider the region $E_1 E_2 E_3^c E_4^c E_5 E_6^c E_7^c E_8^c$.

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- ▶ This region is contained in three single intersections (E_1 , E_2 , and E_5). It's contained in 3 double-intersections ($E_1 E_2$, $E_1 E_5$, and $E_2 E_5$). It's contained in only 1 triple-intersection ($E_1 E_2 E_5$).

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- ▶ Answer: 1. (Follows from binomial expansion of $(1 - 1)^m$.)

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- ▶ How many is that?
- ▶ Answer: 1. (Follows from binomial expansion of $(1 - 1)^m$.)
- ▶ Thus each region in $E_1 \cup \dots \cup E_n$ is counted exactly once in the inclusion exclusion sum, which implies the identity.

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- ▶ There are $\binom{n}{r}$ terms like that in the inclusion exclusion sum. What is $\binom{n}{r} \frac{(n-r)!}{n!}$?
- ▶ Answer: $\frac{1}{r!}$.
- ▶ $P(\cup_{i=1}^n E_i) = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots \pm \frac{1}{n!}$

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- ▶ $P(\cup_{i=1}^n E_i) = 1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \dots \pm \frac{1}{n!}$
- ▶ $1 - P(\cup_{i=1}^n E_i) = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots \pm \frac{1}{n!} \approx 1/e \approx .36788$